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A Novel Nonlinear Acceleration Algorithm for Two-Phase Massively Parallel Reservoir Simulation

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Abstract

Numerical reservoir simulation is a physics-based, high-fidelity approach, and it's critical for optimizing subsurface flow in complex engineering applications such as hydrocarbon recovery, unconventional reservoir development, and CO_2 sequestration, etc. Realistic full-field models often pose numerical challenges due to nonlinearities, phase transitions, and highly heterogeneous media. This work presents a novel nonlinear acceleration method that improves robustness and efficiency of Newton-based solvers by addressing convergence issues arising from derivative discontinuities in the solution space. The proposed approach uses linear system solutions to guide mass updates and restricts updates along convergent directions, effectively managing multiphase transitions and saturated/undersaturated states. Extensive testing on full-field models in production—including those with billions of cells, fine-scale heterogeneity, and diverse rock types—demonstrates that this method consistently achieves 2X–9X speedups over conventional approaches. By directly targeting the root causes of nonlinear solver failure, the method advances the reliability and performance of large-scale reservoir simulations and has reached a level of maturity necessary for routine practical use.

Introduction

Reservoir simulation for giant oil and gas fields—often involving hundreds to thousands of wells and simulation periods spanning several decades or even centuries—remains an extremely time-consuming task, even with the availability of current high-performance computing infrastructure. The computational cost increases significantly with model size and complexity, particularly during history matching and long-term forecasting. Accelerating reservoir simulation performance is therefore critical. Improved computational efficiency enables reservoir engineers to explore a larger number of optimization scenarios within the same time frame, enhancing the decision-making process. This increased simulation throughput can lead to better reservoir management strategies and more optimized hydrocarbon recovery over the asset's life.

In field-scale reservoir simulations of large, heterogeneous, and often fractured oil and gas fields, the solution domain frequently comprises of several millions—or even billions—of grid cells (finite volumes). The total simulation period, including both history matching and prediction phases, can span hundreds of years. Within these models, connate and irreducible saturations vary on a cell-by-cell basis, resulting in highly complex, spatially heterogeneous saturation functions. To honor the original oil in place (OOIP), drainage capillary pressure is typically specified to achieve vertical equilibrium as the initial condition. During simulation, as water saturation increases, capillary pressure is reduced according to the imbibition behavior. In addition, time-varying production and injection well controls, as well as well placements within the domain, introduce another layer of complexity. They act as time-dependent, nonlinear boundary conditions and require fully implicit treatment within the reservoir simulation framework to ensure convergence efficiency (Zhou, 2019). All these nonlinear behaviors are intricate and must be robustly handled by the nonlinear solution system.

For the Newtonian iterative method to perform effectively in solving nonlinear systems, the derivatives of the nonlinear functions must be continuous. However, this continuity is not always present in multiphase, multicomponent flow through porous media. One notable discontinuity arises during fluid phase transitions between saturated and undersaturated states; another occurs when a phase transitions between mobile and immobile states. Each Newtonian iteration searches for a new state (Newton update vector) to advance the solution process. For complex nonlinear functions, Newton's method operates only within a fidelity neighborhood—where the linear approximation remains valid. The method described in this paper strategically selects update vectors to ensure each step remains within this neighborhood. It includes procedures to handle derivative discontinuities, particularly during transitions across phase mobility thresholds and saturation boundaries.

In recent years, researchers have conducted many studies on improvements in nonlinear methods to enhance reservoir simulation performance. Jenny et al. (2006) and Lee et al. (2023) provide comparative studies between sequential fully implicit (SFI) methods—which decouple pressure and transport equations using nested Newton iterations—and fully coupled approaches (Jenny et al. 2006, Lee et al. 2023). Jiang and Tchelepi further discussed several nonlinear acceleration techniques for improving SFI (Jiang and Tchelepi, 2019). While SFI methods offer advantages in certain settings, they can encounter convergence challenges in multiphase reservoir simulations, particularly where strong phase interdependencies and capillary-driven counter-current flows are present. In contrast, the method in this paper solves all equations simultaneously in a fully coupled manner, offering enhanced robustness and stability under such conditions.

Our approach also extends prior work by directly addressing derivative discontinuities that arise during phase transitions at bubble-point or dew-point pressures—areas that have proven challenging for existing methods. For example, the adaptive hybrid strategy proposed by Lu and Beckner applies implicit treatment selectively to non-converged regions following an initial low-implicitness calculation (Lu and Beckner, 2015). While effective in some contexts, this technique may be constrained by time-step limitations due to variable Courant-Friedrichs-Lewy (CFL) numbers and does not explicitly address discontinuities associated with phase transitions. Our method addresses these limitations by offering unconditional stability and flexible time-stepping, while also improving parallel efficiency through better load balancing and avoiding the cache inefficiencies associated with dynamic implicit regions.

Younis's adaptively localized continuation-Newton method is an innovative contribution that assumes failed Newton iterations may still yield valid results for some positive time step (Younis et al, 2009). However, based on our testing in field-scale reservoir simulations, we have observed that this assumption does not always hold. In many practical cases, convergence failures necessitate a time-step reduction. Our method includes targeted procedures for handling derivative discontinuities—frequently the underlying cause of Newtonian divergence—thereby reducing iteration failures and improving overall simulation performance.

Fundamentally, our method targets the root cause of Newton's method failures in reservoir simulation: derivative discontinuities in the solution space. These challenges are especially pronounced in fine-scale, full-field models, where each cell possesses distinct rock properties shaped by geological variability and multiphase rock-fluid interactions. Such nonlinearities lead to diverse saturation functions and end-points across the domain, with phase states often changing between iterations. Existing methods fail to account for these complexities, frequently leading to convergence failures. In contrast, our approach successfully simulates large-scale models with billions of grid cells while preserving full nonlinear complexity. Moreover, it delivers superior material balance convergence—a critical metric for high-fidelity simulations used in reservoir planning and decision-making.

Methodology

This paper discusses a method to accelerate the convergence of the nonlinear systems of equations arising from multiphase flow in porous media. We aim to solve the system using a fully-implicit method in time and finite-volume method in space, wherein the solution domain is subdivided in grid cells, known as finite volumes. The proposed nonlinear acceleration method is designed to be compatible with the massively parallel simulation framework developed in the previous work, ensuring scalability for large-scale reservoir models (Dogru et al. 2009, Fung et al. 2008). The discrete material balance equation for fluid component i for each grid cell are shown in Eqns. 1a and 1b below.

$$\frac{V\phi}{\Delta t}\Delta\left[\rho_o S_o\right] = \Sigma_j T \rho_o \frac{k_{ro}}{\mu_o}\left(\Delta\Phi_o\right) - q_o \tag{1a}$$

$$\frac{V\phi}{\Delta t}\Delta\left[\rho_w S_w\right] = \Sigma_j T \rho_w \frac{k_{rw}}{\mu_w}\left(\Delta\Phi_w\right) - q_w \tag{1b}$$

where k_r is relative permeability; T is phase saturation; T is transmissibility; V is cell volume; Δt is time step size; ρ is molar density; ϕ is porosity; Φ is potential; μ is viscosity; and q_i represents the well term for component i.

In the classical black-oil system, three fluid components—oil, gas, and water—exist within three potential phases. However, in the specific fluid system under consideration in this paper, pressure consistently remains above the bubble point pressure. Consequently, gas exists exclusively in dissolved form within the oil phase rather than as free gas. This results in a simplified two-phase system consisting only of the oleic phase and the aqueous phase. The data input is usually in terms of fluid PVT tables and the parameters for each grid cell include formation volume factors: B_o, B_w; phase viscosities: μ_o, μ_w; rock, undersaturated oil, water compressibility: c_r, c_o, c_w.

Additionally, we have the saturation constraint, and capillary pressure relationship as follows:

$$S_o + S_w = 1 \tag{2}$$

$$P_{cow} = p_o - p_w = f\left(S_w\right) \tag{3}$$

In Eqns. 1a and 1b, relative permeabilities and capillary pressures are typically data input as two-phase rock-fluid tables. They may be experimentally determined or sometimes evaluated empirically using established models. These functions have discontinuous derivatives at their connate and irreducible saturations, S_{wc}, S_{orw}. In simulation practice, each grid cell has unique connate and irreducible saturation values which lead to a different set of rock-fluid tables for each grid cell. The initialization capillary pressure is drainage which must be consistent with measured water saturation and reservoir OOIP values. As water saturation, S_w, increases during simulation, the capillary pressure, P_{cow}, will decrease in a water displacing oil scenario, this will be consistent with an imbibition capillarity.

Figure 1 illustrates the flow charts of method steps in this study. We employ a fully-implicit method in time and finite volume method in space to obtain the discrete material balance equations 1a and 1b. We linearize the coupled nonlinear system of equations via Newtonian iterative method. In residual form, the balance equation for component i in a grid cell has the form shown in Eqn. 4:

$$R_i^{n+1} = \frac{1}{\Delta t}\left(mass_i^{n+1} - mass_i^n\right) + \Sigma_{j \in ngbrs} Flux_i^{n+1} + Q_i^{n+1} \tag{4}$$

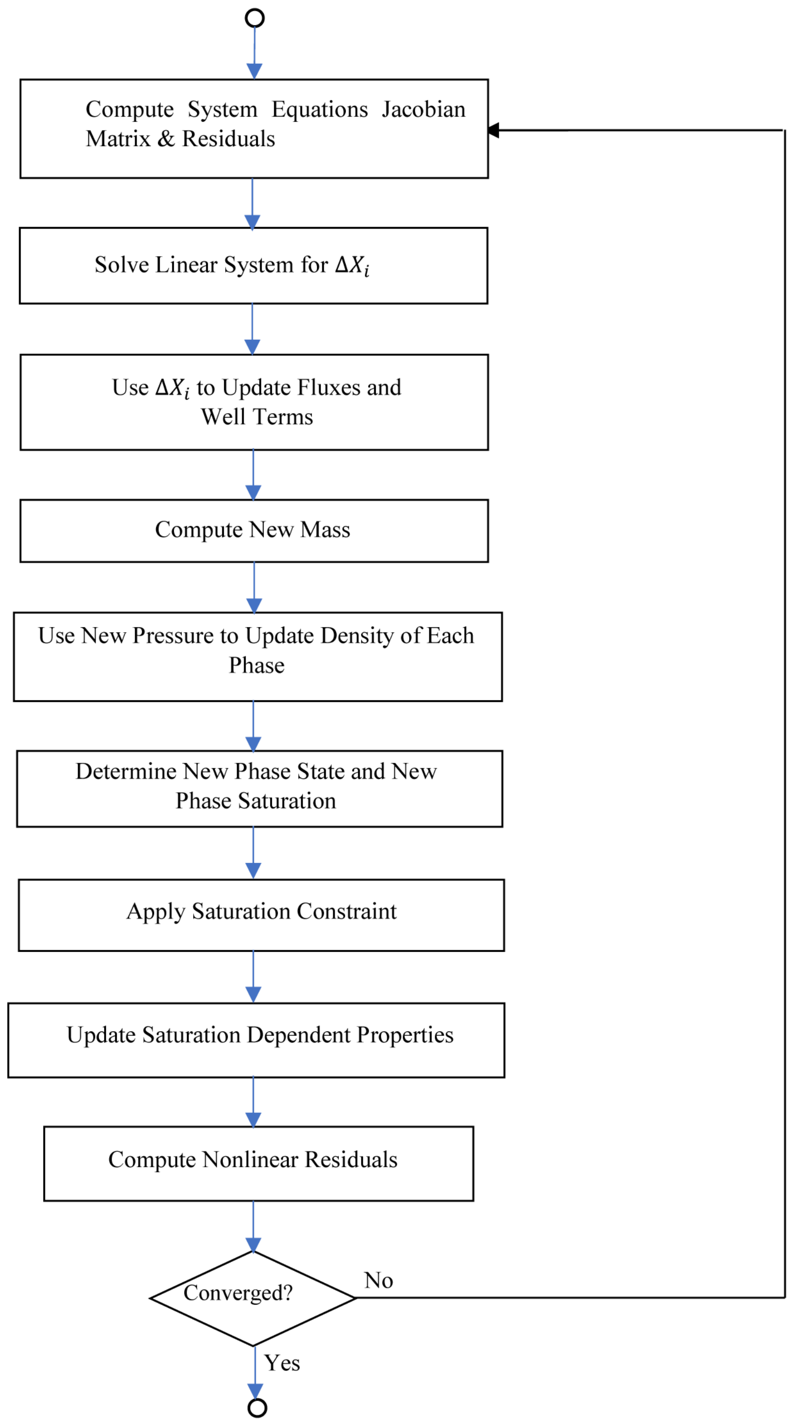

Figure 1—Flow chart illustrating the nonlinear method system

wherein n+1 refers to the new time level. For the k^th^ Newtonian iteration:

$$R_i^{n+1,k+1} \sim R_i^{n+1,k} + \left[\frac{\partial R_i}{\partial X_j}\right]^{n+1,k} * \Delta X_i^k = 0 \tag{5}$$

wherein $\left[\frac{\partial R}{\partial X}\right]^{n+1,k}$ is the Jacobian matrix J.

To find the linear update ΔX_i^k for the primary variables, X_i, for the k^{th} Newtonian iteration, we solve the linear system as in Eqn. 6.

$$\Delta X_i^k = -J^{-1} R_i^{n+1,k} \tag{6}$$

Wherein ΔX_i^k is the linear solution update for the primary variables.

For Newtonian iterative method to work well for solving nonlinear system of equations, the derivatives of the nonlinear functions need to be continuous. This is not always true for the multiphase multicomponent flow in porous media such as the reservoir simulation system (Coats et al. 1995). One such discontinuity occurs during fluid phase switching between the saturated state and the undersaturated state. Another one is when a fluid phase crosses the mobile to immobile boundaries. Each Newtonian iteration attempts to find a suitable new state (the Newton update vector) to continue the search for the solution. If the nonlinear function is complex, the Newtonian iteration has a fidelity neighborhood where the gradient approximation for the solution direction is valid. This paper method selects solution update vector such that each update stays within the solution neighborhood. We update the inter-cell fluxes and well layer flow in Eqn. 4 using the formulae in Eqns. 7a and 7b:

$$Flux_i^{k+1} = Flux_i^k + \left[\frac{\partial Flux_i^k}{\partial X_j}\right]\left\{\Delta X_j\right\} \tag{7a}$$

$$Q_i^{k+1} = Q_i^k + \left[\frac{\partial Q_i^k}{\partial X_j}\right]\left\{\Delta X_j\right\} \tag{7b}$$

Using the mass balance Eqn. 4, and adsorbing $R_i^{n+1,k+1}$ into the new mass updates, we calculate the mass from the old mass and the new fluxes and well flows computed from Eqn. 7b as:

$$Mass_i^{k+1} = Mass_i^n + \Delta t * \left(\Sigma Flux_i^{k+1} + Q_i^{k+1}\right) \tag{8}$$

If hydrocarbon components do not partition into the aqueous phase, the new water saturation, S_w, can be calculated directly:

$$S_w^{k+1} = \frac{Mass_w^{k+1}}{V \emptyset \rho_w^{k+1}} \tag{9}$$

Similarly, the new oil saturation, S_o, can be calculated directly by:

$$S_o^{k+1} = \frac{Mass_o^{k+1}}{V \emptyset \rho_o^{k+1}} \tag{10}$$

We compute the saturation update error as:

$$\varepsilon_S = 1 - S_o^{k+1} - S_w^{k+1} \tag{11}$$

If the saturation error is smaller than the specified tolerance, $\varepsilon_S < tol_S$, the calculated saturation for the grid cell is accepted as the new saturation. Then the saturation dependent properties such as relative permeabilities and capillary pressures are updated. The new nonlinear residuals, $R_i^{n+1,k+1}$, for each component of each cell can now be computed using Eqn. 5. The root mean square (RMS) of the nonlinear residual vector for each fluid component and the overall RMS residual vector of all components are calculated. They are compared against input residual tolerances to determine whether acceptable convergence has been reached. We also compute the total material balance error for each fluid component for the time step and evaluate the error in terms of a fraction of the original material in the simulation domain. If the estimated total number of timestep times the fraction is not smaller than an acceptable global accumulated material balance error tolerance, convergence is also considered not achieved. If convergence has not been achieved, nonlinear iteration is advanced to $k + 1$, and program control goes back. Otherwise, the current time step has converged and program control advances to the next time step.

Applications

We have successfully tested the paper method in a wide range of field-scale simulation models and demonstrated the substantial improvement in efficacy over prior art method. Comparing to prior art method, for the same model, the method is able to achieve nonlinear convergence with larger time steps and with fewer iterations. It also has far fewer time-step cuts and achieve significant computer time savings. The new method is shown to have better material balance accuracy and at a better convergence rate.

The demonstration model has 50 million cell corner-point geometry (CPG), irregular structured grid. A picture of the reservoir model is shown in Figure 2. It has several rock-fluid regions as well as multiple equilibrium regions with tilted oil-water contacts. Connate and irreducible saturations are defined on a cell-by-cell basis, allowing for variations of relative permeabilities and capillary pressure. This simulation model has thousands of wells.

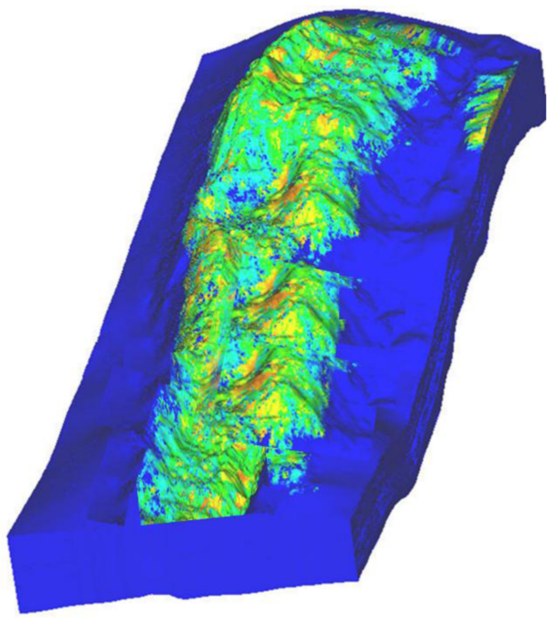

Figure 2—An oil-water two-phase, 50 million cell reservoir simulation model for testing

Using the prior art method, the model required significantly more computational effort and time. In contrast, the new method delivers a 3.41X speedup in wall time, along with a 4.35X reduction in the number of time steps, and a 23.35X reduction in time-step cuts. The nonlinear and linear solver iterations are also reduced by 3.84X and 2.20X, respectively. Additionally, the average time-step size is increased by 4.27X, reflecting improved numerical stability and efficiency. These improvements are achieved without compromising accuracy, as the material balance errors remain comparable or are slightly improved in the new method. The final material balance errors of the new method are at Oil = 3.153×10^{-5}, and Water = 1.653×10^{-5}, while the final material balance errors are at Oil = 4.086×10^{-4}, and Water = 1.343×10^{-5} for the prior art method. Units of the material balance errors are unitless. Both model runs were conducted on the same HPC clusters using the same amount of hardware resources. Figure 3 demonstrates the comparison of simulation results: the current method versus prior-art method. The method gives the same results, but numerical performance is significantly better as shown in Figure 4. Figure 5 shows the oil material balance error comparision along the simulation process. Details on the new method's acceleration metrics are listed in Table 1. These results demonstrate the efficacy the new method, which gives the same production results, and at a better material balance convergence, but at a fraction of the computational cost of the prior art method.

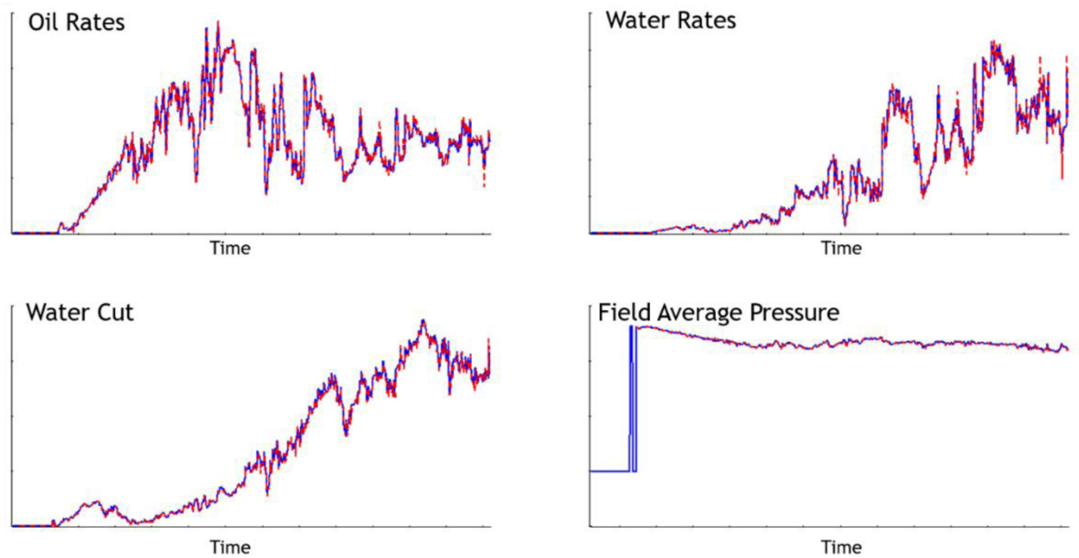

Figure 3—Comparison of simulation result: the new method (blue solid curve) versus prior-art method (red dashed curve). New method gives the same result.

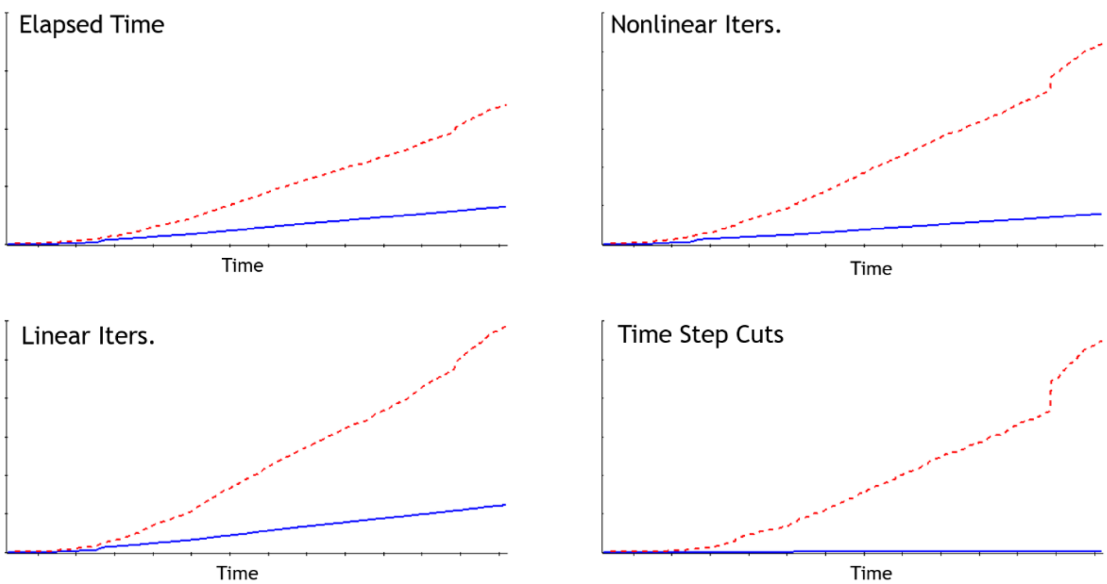

Figure 4—Comparison of running performance: this new method (blue solid curve) versus prior-art method (red dashed curve). New method gives a theoretical speedup factor of approximately 4 times.

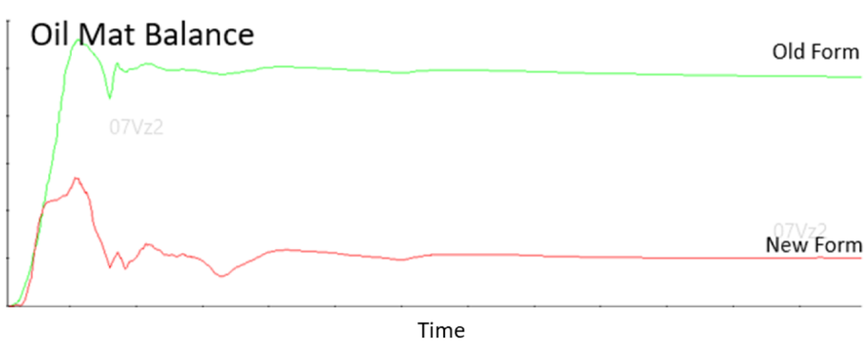

Figure 5—Comparison of running material balance: performance improvements are achieved without compromising accuracy

Table 1—An oil-water two-phase, 50 million cell reservoir simulation model performance improvement ratio

	Improvement Ratios
Wall Time	3.41X
No. of Time Steps	4.35X
No. of Time Step Cuts	23.35X
No. of Nonlinear Iterations.	3.84X
No. of Linear Iterations	2.20X
Avg. Time Step Size	4.27X

For test model, we have also run the 1-billion-cell (3*3*2 refinement) fine grid model and the 9.5-billion-cell (9*9*2 refinement) fine-grid models with history and prediction period totaling hundreds of years and find the method to be stable and robust. Prior-art method shows difficulties running models at these fine scales.

The application of the new nonlinear method across a diverse set of reservoir simulation models has demonstrated consistent and substantial improvements in computational performance, as summarized in Table 2. These models, representing different geological and operational scenarios from field applications, vary significantly in size and duration, yet all exhibit notable reductions in solution time and total wall-clock runtime. For example, models resulted in improvement factors ranging from 1.80X to as high as 9.15X in running time. These simulations were executed utilizing hundreds to thousands of massively parallel cores on HPC cluster, which is manufactured by HPE, featuring AMD 7702 cores and Slingshot-10 interconnect. These gains are attributed to nonlinear path optimization, showcasing the broad applicability and robustness varying reservoir conditions and model complexities, and the new method has reached a level of maturity necessary for routine practical use.

Table 2—A Diverse Set of Reserovir Simulation Models with Improvement Factors

Model	Model Sizes	Simulation Durations	Improvement Factors of Running Time
Model A	61 Million	HM+PR	1.99X
Model B	62 Million	HM	1.80X
Model C	52 Million	HM+PR	2.65X
Model D	38 Million	HM+PR	2.18X
Model E	66 Million	HM+PR	9.15X
Model F	60 Million	HM+PR	3.11X
Model G	25 Million	HM	2.94X
Model H	25 Million	HM	2.39X
Model I	52 Million	HM	3.05X
Model J	62 Million	HM+PR	3.64X
Model K	60 Million	HM+PR	2.11X

Conclusions

This work presents a robust nonlinear acceleration strategy tailored for large-scale, high-fidelity reservoir simulations involving complex multiphase flow behavior. By identifying and addressing key sources of convergence failure—particularly derivative discontinuities associated with phase transitions and fluid mobility changes—the proposed method enhances the stability and efficiency of fully implicit formulations.

Its integration into a scalable parallel framework enables consistent performance across models with billions of cells with highly heterogeneous properties, and complex nonlinear functions which are different cell-by-cell. Extensive validation on diverse case studies demonstrates significant improvements in convergence rate and computational speed, offering a practical solution for advancing field development planning and subsurface resource management.

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